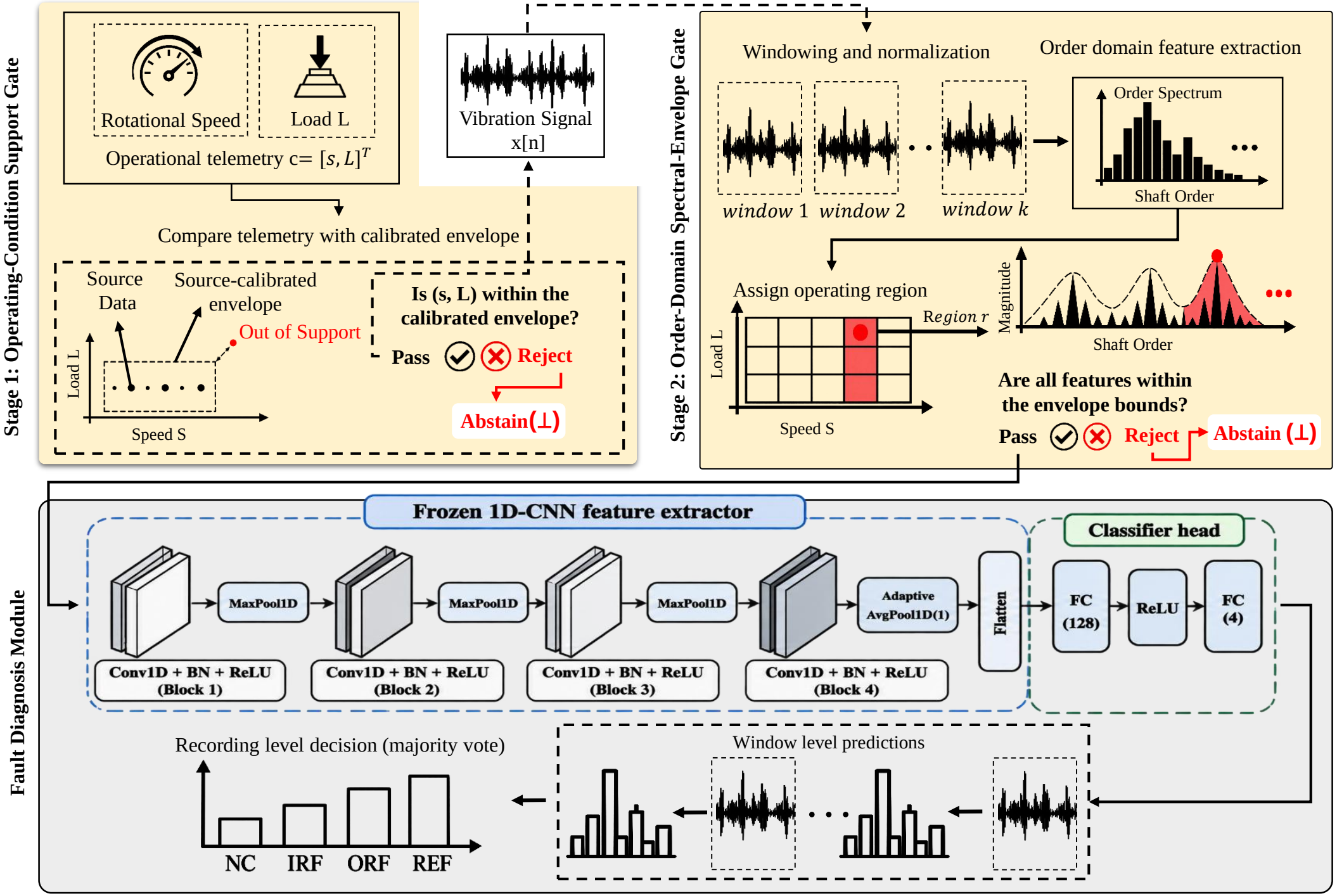

Stage 1: Operating-Condition Support Gate
Rotational Speed
Load L
Operational telemetry c= $[s, L]^T$
Vibration Signal x[n]
Compare telemetry with calibrated envelope
Source Data
Source-calibrated envelope
Out of Support
Load L
Speed S
Is (s, L) within the calibrated envelope?
Pass
Reject
Abstain(⊥)
Stage 2: Order-Domain Spectral-Envelope Gate
Windowing and normalization
Order domain feature extraction
window 1
window 2
window k
Order Spectrum
Shaft Order
Assign operating region
Load L
Speed S
Region r
Magnitude
Shaft Order
Are all features within the envelope bounds?
Pass
Reject
Abstain (⊥)
Fault Diagnosis Module
Frozen 1D-CNN feature extractor
Conv1D + BN + ReLU (Block 1)
MaxPool1D
Conv1D + BN + ReLU (Block 2)
MaxPool1D
Conv1D + BN + ReLU (Block 3)
MaxPool1D
Conv1D + BN + ReLU (Block 4)
Adaptive AvgPool1D(1)
Flatten
Classifier head
FC (128)
ReLU
FC (4)
Window level predictions
Recording level decision (majority vote)
NC
IRF
ORF
REF